



Semester 2 Examinations 2014/ 2015

Exam Code(s)	3BME1, 3BMS2, 3BPP1, 3BS9, 3FM2, 3MR3, 4BME1, 4BMS2, 4BS2
Exam(s)	3 rd & 4 th Arts, 3 rd & 4 th Science
Module Code(s)	MP307, MP341
Module(s)	Modelling II, Mathematical Modelling
Paper No.	1
Repeat Paper	No
External Examiner(s)	Professor P. Martin
Internal Examiner(s)	Dr P. Piironen

Instructions: For full marks answer question 1 in Section A **and** TWO questions in Section B correctly.
All questions carry the same weight.

Duration	2 hours
No. of Pages	6
Discipline(s)	Applied Mathematics
Course Co-ordinator(s)	Dr P. Piironen

Requirements:

Release in Exam Venue	Yes	☒	No	☐
MCQ	Yes	☐	No	☒
Handout	None			
Statistical/ Log Tables	None			
Cambridge Tables	Yes			
Graph Paper	None			
Log Graph Paper	None			
Other Materials	Calculator			
Graphic material in colour	Yes	☐	No	☒

Section A - Answer all questions (a)-(d). (Each question is worth **5 Marks**.)

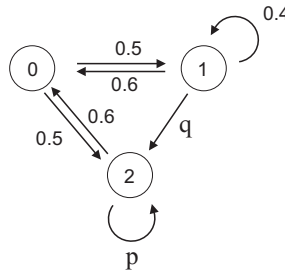
1. (a) Consider an infinite queue and let

$$p_k = \left(\frac{1}{4}\right)^k \frac{3}{4}, \quad k = 0, 1, 2, \dots \quad (\text{Geometric distribution})$$

be the probability that the queue is of length k .

- (i) What is the expected queue length (the mean)?
- (ii) What is the variance and the standard deviation of the queue length?

- (b) Consider the following Markov Process with transition probabilities shown.



- (i) Find the transition matrix P and the probabilities p and q .
 - (ii) Is the Markov process ergodic? Explain your answer.
 - (iii) Find the equilibrium probabilities.
- (c) Consider a model for a non-fatal disease in a population P given by

$$\begin{aligned} \frac{dS}{dt} &= -\alpha SI, \quad S(0) = S_0, \quad \alpha > 0, \\ \frac{dR}{dt} &= \beta I, \quad R(0) = 0, \quad \beta > 0, \end{aligned}$$

where S , I and R respectively correspond to susceptible, recovered and infected groups of the population. Let $\rho = \beta/\alpha$ and show that the relation between the groups S and R can be described by

$$\begin{aligned} S(R) &= S_0 e^{-\frac{R}{\rho}}, \\ R(S) &= -\rho \ln \left(\frac{S}{S_0} \right). \end{aligned}$$

(d) Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= b_1 x_2, \\ \frac{dx_2}{dt} &= x_1 + x_2 + u, \\ y &= x_1 + x_2,\end{aligned}$$

where $x_1 \in \mathbb{R}$ and $x_2 \in \mathbb{R}$ represent the state, $u \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output and $b_1 \in \mathbb{R}$ is the control parameter.

- (i) Is this system observable? Explain your answer.
- (ii) Is this system reachable? Explain your answer.

Section B - Answer any two questions.

2. The size of queue is given by a nearest-neighbour Markov model. It has a Poisson arrival pattern with parameter α_k and a Poisson servicing pattern with parameter β_k , where k is the queue size.

- (a) Let the maximum size of the queue be r .
(i) Explain why $\alpha_r = \beta_0 = 0$.
(ii) Assuming that the Markov Process is ergodic, show that the equilibrium probabilities are given by

$$\pi_k = \rho_k \rho_{k-1} \dots \rho_1 \pi_0$$

for $0 < k \leq r$ where $\rho_k = \alpha_{k-1}/\beta_k$.

8 Marks

- (b) Let the queue be infinite with

$$\alpha_k = \alpha, \quad \beta_k = k\beta,$$

where $\alpha > 0$ and $\beta > 0$.

Show that the equilibrium probabilities π_k , $k = 0, 1, 2, \dots$, are given by the Poisson distribution

$$\pi_k = \frac{\rho^k e^{-\rho}}{k!}, \quad \rho = \frac{\alpha}{\beta}.$$

6 Marks

- (c) Let the queue correspond to a call centre with 3 agents. Each agent can deal with 1 customer enquiry per 3 minutes on average. If 20 calls per hour arrive on average what is the probability that the call centre queue is saturated?

6 Marks

3. Consider the nonlinear population model for a population P given by

$$P_{n+1} = R(P_n)P_n,$$

where

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}$$

and $r, K > 0$ are constants.

- (a) Discuss why the function $R(P)$ is important for determining if the population will grow or decline.

4 Marks

- (b) Find the fixed points for this model.

6 Marks

- (c) Let $r = 2$, $K = 10$ and describe the long-term behaviour of the population P if

- i. $P_0 = 1$,
- ii. $P_0 = 100$,
- iii. $P_0 = 200$.

10 Marks

4. A two-compartment model for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -p_0 - p_1 & p_1 \\ p_2 & -p_2 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} b_0 \\ 0 \end{pmatrix} u, \quad y = c_1 + c_2,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, p_0 , p_1 , p_2 and b_0 are rate constants, u is the flow rate of the drug into compartment 1 and y is the sum of the concentrations of the drug in compartments 1 and 2.

- (a) Draw a schematic diagram of this system including concentrations, flow directions and rate parameters.

4 Marks

- (b) Investigate for what values of p_0 , p_1 and p_2 this system is reachable.

6 Marks

- (c) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $r = 1$ and the closed loop eigenvalues for the system satisfies $\lambda_1 = -3$ and $\lambda_2 = -2$.

10 Marks